Regression

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Introduction

Regression analyses are a set of methods that allow you not only to describe how a variable changes based on other variable, but also to infer and to make predictions. You have many possibilities depending on your data:

- Linear regression (dependent variable is a number)
- Logistic regression (dependent variable is a category - binary)
- Cox regression (dependent variable is time to an event - incidence rate)
- Poisson regression (dependent variable is a count)

The simplest relationship follows the structure of a dependent (y) and independent (x) variables framework.

$$y = \alpha + \beta x + \varepsilon$$

y = dependent variable (response)

α = intercept

β = coefficient

x = independent variable (predictor)

ε = residuals (error)

Linear regression

For more details, please read this paper

Example:

```
# install.packages("lgrdata")
# You should first install this package to load the data that it contains
library(lgrdata) # load the package
library(tidyverse)
data(howell) # load the dataset
str(howell) # see the structure
# ggplot(howell, aes(x=weight, y=height))+
#  geom_point()
```

Fit a linear model between height and weight

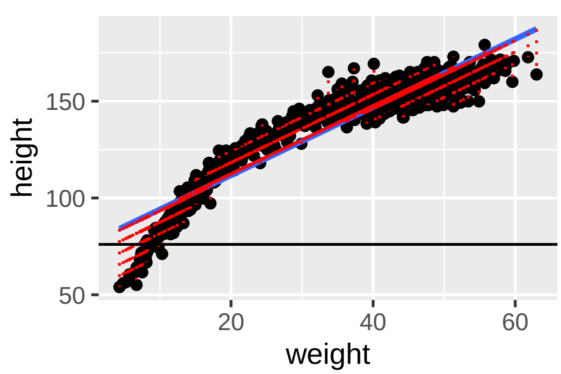
```
# Run a linear model
linear_model <- lm(height ~ weight, data = howell)
summary(linear_model)
```

```
Call:
lm(formula = height ~ weight, data = howell)

Residuals:
    Min      1Q  Median      3Q     Max 
-33.221  -5.786   0.752   6.874  29.285 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  76.5348     0.8814   86.84   <2e-16 ***
weight        1.7591     0.0234   75.17   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.411 on 781 degrees of freedom
Multiple R-squared:  0.8786,	Adjusted R-squared:  0.8784 
F-statistic:  5650 on 1 and 781 DF,  p-value: < 2.2e-16
```

```r
# Run a correlation test
cor.test(howell$weight, howell$height, method = "pearson")
```

```
	Pearson's product-moment correlation

data:  howell$weight and howell$height
t = 75.165, df = 781, p-value < 2.2e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.9282044 0.9452961
sample estimates:
      cor
0.9373115
```

```r
# Make an interpretation
res <- residuals(linear_model)
predicted <- fitted(linear_model)

ggplot(howell, aes(x = weight, y = height)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  geom_segment(aes(x = weight, xend = weight, y = predicted, yend = height),
               color = "red", linetype = "dotted") +
  geom_hline(yintercept = 76)
```



Can you guess now how will be the formula? $y = \alpha + \beta x + \varepsilon$

Can you predict now the height of someone with a weight of 62?

Logistic regression

This analysis is also called generalized linear models (glm). The dependent variable must be 1 or 0.

```
# Read the myeloma dataset and categorize TP53 expression
library(readxl)
myeloma <- read_excel("/Users/dfmoreno/Dropbox/Lectures/myeloma.xlsx")
mean(myeloma$TP53)
```

```
[1] 1537.303
```

```
myeloma <- myeloma %>%
  mutate(low_tp53 = ifelse(TP53<=1500, 1, 0))
table(myeloma$low_tp53)
```

```

  0   1 
116 140 
```

```
# Run a glm to explain/predict low TP53 expression
log_reg <- glm(low_tp53 ~ molecular_group, data = myeloma, family = binomial)
exp(coef(log_reg)) # convert the coefficients to odds ratio (OR)
```

```
                  (Intercept)       molecular_groupCyclin D-2 
                    1.2000000                       1.5555556 
   molecular_groupHyperdiploid molecular_groupLow bone disease 
                    0.4166667                       1.1538462 
           molecular_groupMAF             molecular_groupMMSET 
                    2.0833333                       2.2222222 
  molecular_groupProliferation 
                    0.6770833 
```

How do you interpret the odds ratio?

How can you visualize the discrimination of low_tp53 based on molecular_group?

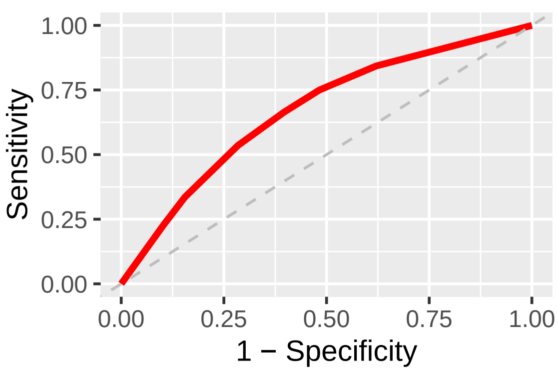
```
# install.packages("pROC")
library(pROC)
# Calculate predicted probabilities
prob <- predict(log_reg, type = "response")

# Generate a ROC object
roc_obj <- roc(myeloma$low_tp53, prob)
ci(roc_obj) # confidence interval
```

```
95% CI: 0.6048-0.7358 (DeLong)
```

```
plot(roc_obj, col = "blue", main = "ROC Curve", print.auc = TRUE)
```

Using a customized plot

```
# Using ggplot
roc_df <- data.frame(
  specificity = roc_obj$specificities,
  sensitivity = roc_obj$sensitivities
)

ggplot(roc_df, aes(x = 1 - specificity, y = sensitivity)) +
  geom_line(color = "red", size = 1.2) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "grey") +
  xlab("1 - Specificity") +
  ylab("Sensitivity")
```



Cox regression

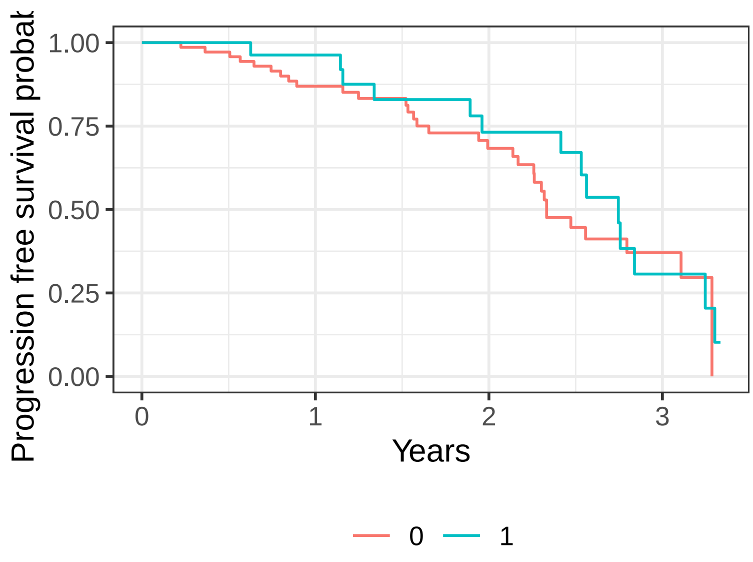
Survival data is usually modeled by Cox regression analyses. They are also called Proportional Hazards Model, because there is an assumption that the hazard ratios should be constant throughout time.

```
# Install the following packages
# install.packages("survival")
# install.packages("survminer")
# install.packages("ggsurvfit")
library(survival)
library(survminer)
library(ggsurvfit)

# Obtain the primary biliary cirrhosis (pbc)
# status (1=liver transplant, 2=death, 0=alive)
# time: time to last follow-up
data(pbc)
head(pbc) # this will get pat of the dataset

# Fit a cox regression model and interpret it
cox_stage1 <- coxph(Surv(time, status == 2) ~ stage, data = pbc)
summary(cox_stage1)

# Fit Cox regression model
cox_stage2 <- coxph(Surv(time, status == 2) ~ factor(stage), data = pbc)

# Estimate survival curves for plotting
survival <- survfit2(Surv(time, status == 2) ~ 1, data = pbc)
ggsurvfit(survival) +
  labs(x = "Days", y = "Progression free survival probability")
```

What are the coefficients in cox_stage2?

```
# What if I need to calculate the time?
# The package lubridate handles dates easily
# install.packages("lubridate")
library(lubridate)
library(tidyverse)

event_data <- read_xlsx("/Users/dfmoreno/Dropbox/Lectures/dates_file.xlsx")
```

```r
event_data <- event_data %>%
  mutate(pfs_yrs = as.duration(dx_date %--% event_date) / dyears(1))

# Also try
# mutate(pfs_yrs = time_length(dx_date %--% event_date, unit = "years"))
# mutate(pfs_yrs = time_length(dx_date %--% event_date, unit = "months"))
# mutate(pfs_yrs = time_length(dx_date %--% event_date, unit = "days"))

survival_data <- survfit2(Surv(pfs_yrs, status==1) ~ protein1, data = event_data)

ggsurvfit(survival_data) +
  labs(x = "Years", y = "Progression free survival probability")
```




```r
# Obtain the log-rank test using survdiff (from survival package)
survdiff(Surv(pfs_yrs, status==1) ~ protein1, data = event_data)
```

```
Call:
survdiff(formula = Surv(pfs_yrs, status == 1) ~ protein1, data = event_data)

             N Observed Expected (O-E)^2/E (O-E)^2/V
protein1=0 71       31     28.6     0.208      0.61
protein1=1 29       14     16.4     0.362      0.61

 Chisq= 0.6  on 1 degrees of freedom, p= 0.4
```

Multivariable regression models

So far, we covered univariable regression models to explain or predict a certain outcome. However, an outcome or event are usually explained by a more complex relationship of independent variables (predictors). For example, how to predict good or bad prognosis in a certain cancer based on the combination of expression of RNA, proteins and the presence of mutations.

Fortunately, you just need to adjust for the other covariates in the same formula.

```r
# You can add more variables
glm(low_tp53 ~ molecular_group + chr1q21_status, data = myeloma, family = binomial) %>%
  summary()
```

```
Call:
glm(formula = low_tp53 ~ molecular_group + chr1q21_status, family = binomial,
    data = myeloma)

Coefficients:
                                Estimate Std. Error z value Pr(>|z|)
(Intercept)                     -0.27485    0.53714  -0.512    0.609
molecular_groupCyclin D-2        1.04103    0.64467   1.615    0.106
molecular_groupHyperdiploid     -0.96801    0.64030  -1.512    0.131
molecular_groupLow bone disease  0.07574    0.69775   0.109    0.914
molecular_groupMAF               0.59963    0.82980   0.723    0.470
molecular_groupMMSET             0.85972    0.66478   1.293    0.196
molecular_groupProliferation    -0.21127    0.69239  -0.305    0.760
chr1q21_status3 copies           0.47341    0.42939   1.103    0.270
chr1q21_status4+ copies          0.34363    0.49851   0.689    0.491

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 257.51  on 185  degrees of freedom
Residual deviance: 229.43  on 177  degrees of freedom
  (70 observations deleted due to missingness)
AIC: 247.43

Number of Fisher Scoring iterations: 4
```